

Project 5: R&D Mode - Voice AI Agent Evaluation

Project 5.1: Deepgram, Eleven Labs, & Vercel Agentic Voice Platform Evaluation

Description

This **six-week** research and development project aims to technically evaluate and compare two leading voice AI platforms (Deepgram and Eleven Labs) for building a modern, full-stack, agentic voice platform. The evaluation will focus on Deepgram AI's Flux model for low-latency Speech-to-Text (STT) and both Deepgram's and Eleven Labs Voice AI's Text-to-Speech (TTS) capabilities. A core component of this project is to explore and implement integration patterns with the **Vercel AI SDK (version 6)** in a **Next.js/TypeScript** environment, ensuring the final, working prototype is designed for deployment on **Vercel Cloud**. The goal is to determine the optimal platform and architecture based on performance, agentic feature set, cost, and applicability for high-signal conversational AI.

Outline

I. Core Evaluation (Must Have)

- **Speech-to-Text (STT) Benchmark:** Evaluate Deepgram AI's Flux model performance and latency. Compare pricing against OpenAI's Realtime voice API (technical comparison is out of scope).
- **Text-to-Speech (TTS) Comparison:** Evaluate and compare Deepgram's TTS and Eleven Labs Voice AI on key metrics: voice quality, naturalness, and feature parity (e.g., low-latency streaming).
- **Agentic Capabilities & Integration:** Compare the ease of integration for building complex agents, focusing on:
 - Native support for RAG and function/tool calling.
 - Seamless integration with the **Vercel AI SDK (v6)** and Next.js/TypeScript architecture.

II. Deliverables

- **Working Prototype:** A functional, end-to-end voice agent prototype integrated with STT/TTS, RAG, and function tool calls, built on Next.js/TypeScript and deployable on Vercel Cloud.
- **Cost & Applicability Analysis:** A detailed comparison of pricing, licensing, and overall suitability for both voice agent and telephony agent use cases.
- **Summary Research Report:** A comprehensive report documenting all findings, technical specifications, and a final architectural recommendation.

- **Open-Sourced Code Repository:** All prototype code, benchmarks, and configuration will be published to a public repository for community reference.

III. 6-Week Project Plan (High-Level Phased Approach)

Week	Phase	Focus
Week 1	Foundational Setup & STT Benchmark	Setup API access and core Next.js/TypeScript project environment. Conduct Deepgram Flux STT Benchmark (performance, latency, and OpenAI pricing comparison). Develop basic voice I/O loop prototype.
Week 2	TTS Comparison & Vercel AI SDK Integration	Conduct TTS Comparison (Deepgram vs. Eleven Labs quality and features). Begin integrating the chosen TTS and STT into the Next.js prototype using the Vercel AI SDK (v6) .
Week 3	Agentic Feature Integration	Finalize the Working Prototype by integrating the agent with RAG and function tool calls to test the platforms' Agentic Capabilities and full-stack flow.
Week 4	Testing & Cost Analysis	Perform final end-to-end testing of the full voice agent prototype on a Vercel Cloud deployment . Complete the Cost & Applicability Analysis for all tested platforms.
Week 5	Documentation & Report Drafting	Draft the comprehensive Summary Research Report . Finalize the documentation, code cleanup, and commenting for the open-source repository.

Week 6	Project Finalization & Publishing	Final security and performance review. Project Presentation/Sign-off. Publish the Summary Research Report and the Open-Sourced Code Repository . Project Closure and handover.
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IV. Resourcing and Dependencies

- **Project Manager:** Callum Bir
- **Dependencies:** Access to Deepgram and Eleven Labs APIs for testing. Stability of the Vercel AI SDK (v6) and Vercel Cloud platform.